Heart Disease Prediction

Comparative Evaluation of Machine Learning Models:

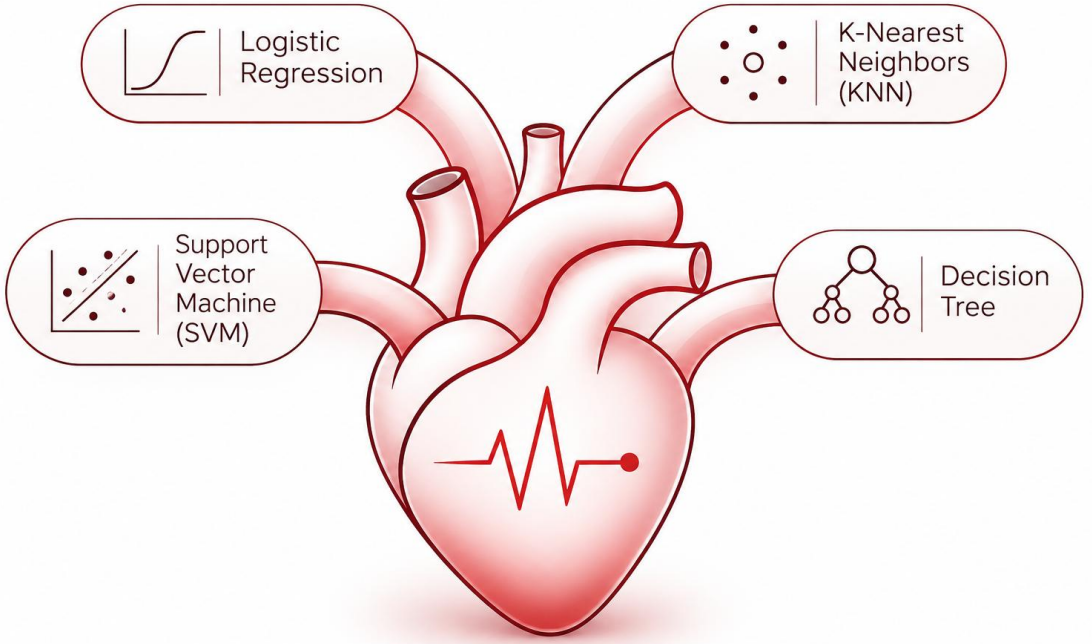


Image generated with AI

Presented by:

Carlos Alvarez, Felipe Kuerten Neves & Karina Adorno

November 13th 2026

The problem

Heart disease is the leading cause of death in the world.

Early detection can save lives.

Goal

Predict if a patient has heart disease or not, using clinical data.

Type of task

Binary classification:

0 = No Disease,

1 = Heart Disease.

Why machine learning

Help doctors identify patients at risk faster.

Data Lineage

1988
UCI REPOSITORY

Heart Disease collection donated to UCI.

76 raw attributes

1989
ORIGINAL CLINICAL PAPER

Detrano et al.

- 303 Cleveland
- 425 Hungary
- 200 Long Beach
- 143 Switzerland
- Total = 1,071 patients.

2026
ML COMPARATIVE STUDY

Aldila & Supriyono studied Cleveland only.

- Logistic regression
- SVM
- KNN
- Decision tree

Aldila, A. S., & Supriyono, L. A. (2026). *Evaluating Logistic Regression, SVM, KNN, and Ensemble Models for Accurate Heart Disease Risk Prediction*. JITK, 11(3), 949–957.

The dataset: Cleveland Heart Disease (UCI)

Feature	Meaning
age	Age in years
sex	0 = female, 1 = male
cp	Chest pain type (4 types)
trestbps	Resting blood pressure
chol	Cholesterol
fbs	Fasting blood sugar > 120
restecg	Resting ECG result

Feature	Meaning
thalach	Maximum heart rate
exang	Chest pain during exercise
oldpeak	ST depression (ECG)
slope	Slope of the ST segment
ca	Number of major vessels (0–3)
thal	Thalassemia test result
num → target	Diagnosis (0–4 → 0 / 1)

303 patients · 13 features + 1 target

Data cleaning

Missing values

6 patients had missing values in **ca** or **thal**. As in the paper, they were removed.

303 → 297

Binary target

The original diagnosis goes from 0 to 4. We grouped 1–4 as "disease".

0 / 1

Class balance after cleaning

No Disease: 160 (54%) | **Heart Disease: 137 (46%)**

What the data shows

Mostly men

About 68% of the patients are male. The model may work less well for women.

Outliers kept

They are clinically possible, so we kept them, as the paper did.

Weak features

Cholesterol and blood sugar have almost no relation with the diagnosis.

Strongest relationships with heart disease

Feature	Correlation
thal (thalassemia test)	+0.53
ca (major vessels)	+0.46
oldpeak (ST depression)	+0.42
exang (exercise angina)	+0.42
cp (chest pain type)	+0.41
thalach (max heart rate)	−0.42

PART 2 · FELIPE

Our approach

1
Clean
297 patients

2
Split 80 / 20
stratified

3
Scale
range 0–10

4
Train
paper’s settings

5
Tune
6-fold CV

6
Evaluate
on the test set

Primary metric: recall

PART 2 · FELIPE

Pipeline: no data leakage

Encoding

Categories → codes 0, 1, 2...

Scaling 0–10

Same range for all features

Learns from train data

Test data is used only at the end

```
def make_pipeline(model):
  return Pipeline([
    ("encoder", clone(encoder)),
    ("scaler", MinMaxScaler(
      feature_range=(0, 10))),
    ("model", model),
  ])
```

Same pipeline for every model, so the comparison is fair.

PART 2 · FELIPE

The four models

Logistic Regression

Linear equation → probability

Paper: C = 0.1

KNN

Vote of the K nearest patients

Paper: K = 7

SVM

Best boundary between classes

Paper: RBF, C = 3, gamma = 0.65

Decision Tree

A series of yes / no questions

Paper: random_state = 5

In the paper, the KNN and SVM settings are swapped. We used each one with the correct model.

PART 2 · FELIPE

Reproducing the paper

We copied the paper's settings and ran them on our test set (60 patients, 28 sick).

Model	Paper accuracy	Our accuracy	Sick we found	Result
Logistic Regression	86.9%	83.3%	22 of 28	Close
KNN (K = 7)	88.5%	86.7%	23 of 28	Close
Decision Tree	85.3%	73.3%	19 of 28	Overfits
SVM	86.9%	53.3%	**1 of 28**	Fails

> Same settings, very different SVM: ours says **"healthy" to almost everyone**.

PART 2 · FELIPE

Why the paper’s SVM settings fail

The SVM compares patients. **Gamma** sets how strict the comparison is.

PAPER’S SETTINGS · gamma = 0.65

Too strict

Every patient looks unique, so it says “healthy” to almost everyone.

1 of 28

sick patients found

OUR SETTINGS · gamma = 0.001

Just right

Similar patients look similar, so it learns the patterns.

22 of 28

sick patients found

> **The SVM is fine. The gamma written in the paper is the problem.**

PART 2 · FELIPE

Hyperparameter tuning

Model	Settings tested
Logistic Regression	C
KNN	K, weights, metric
SVM	kernel, C, gamma
Decision Tree	depth, min samples

GridSearchCV tests every combination with stratified 6-fold cross-validation, as in the paper.

```
search = GridSearchCV(
  make_pipeline(model),
  param_grids[name],
  cv=cv,
  scoring=SCORING,
  refit="recall",
)
search.fit(X_train, y_train)
```

Chosen by **recall**, on training data only.

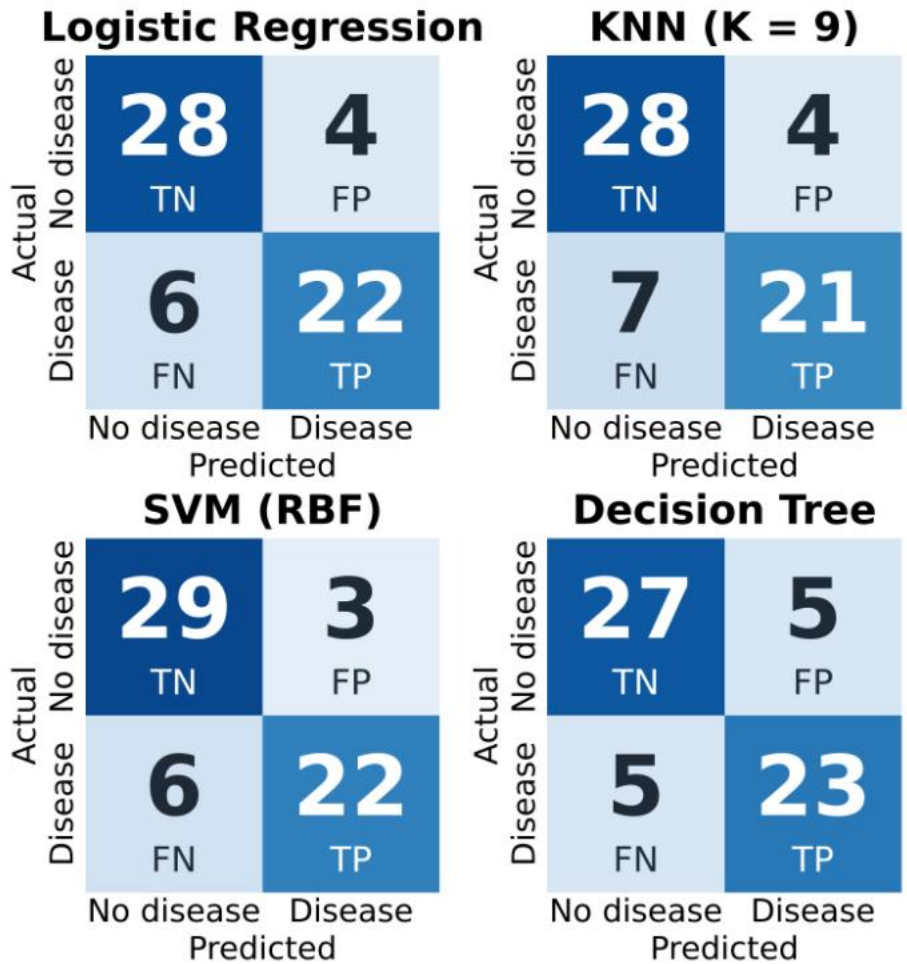
PART 2 · FELIPE

Results after tuning

Test set: 60 patients (28 sick). Missed = sick patients predicted as healthy.

Model	Settings	Accuracy	Recall	AUC	Missed
Logistic Regression	C = 0.01	83.3%	0.79	0.95	6
KNN	K = 9	81.7%	0.75	0.94	7
SVM	gamma 0.001	85.0%	0.79	0.96	6
Decision Tree	depth = 3	83.3%	0.82	0.87	5

> Only **1 to 3 patients** apart: no model is clearly superior.



PART 3 · CARLOS

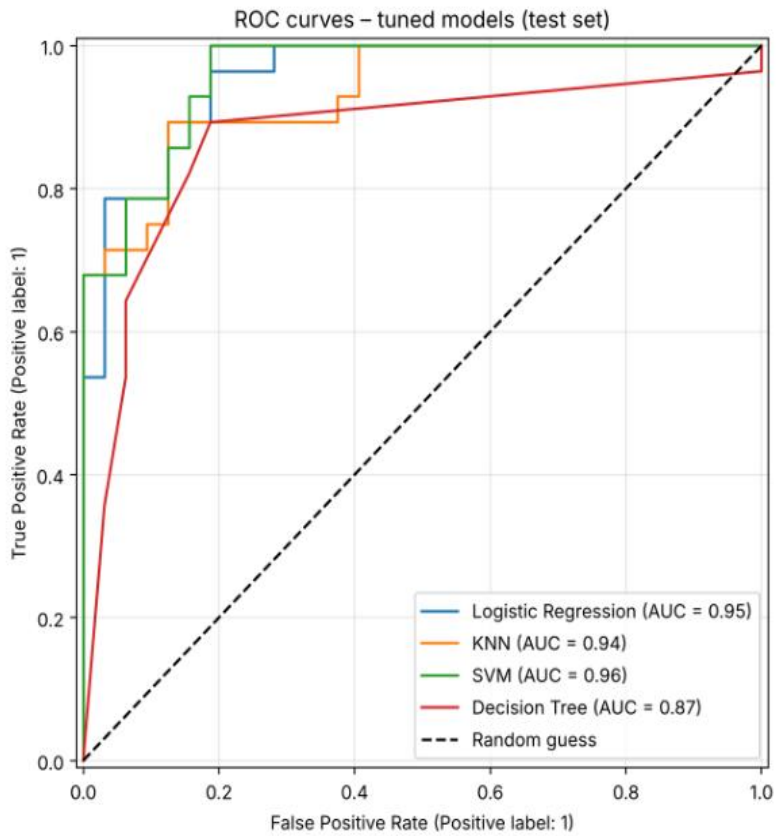
Confusion matrices

Tuned models · 60 test patients

Code	Meaning
TN	Healthy, predicted healthy
FP	Healthy, predicted sick
FN	Sick, predicted healthy
TP	Sick, predicted sick

> **FN is the worst error.**
> Tree 5 · LR 6 · SVM 6 · KNN 7

ROC curves and AUC



AUC = the chance that a sick patient gets a higher risk score than a healthy one.

Model	ROC-AUC
SVM	0.96
Logistic Regression	0.95
KNN	0.94
Decision Tree	0.87

1.0 = perfect · 0.5 = random guess (dashed line)

PART 3 · CARLOS

Comparison with the paper

Each cell: **paper / ours** (tuned models)

Model	Accuracy	Recall	Precision	F1	AUC
LR	86.9% / 83.3%	0.84 / 0.79	0.90 / 0.85	0.87 / 0.81	0.92 / 0.95
KNN	88.5% / 81.7%	0.72 / 0.75	0.92 / 0.84	0.81 / 0.79	0.92 / 0.94
SVM	86.9% / 85.0%	0.84 / 0.79	0.90 / 0.88	0.87 / 0.83	0.93 / 0.96
Tree	85.3% / 83.3%	0.78 / 0.82	0.93 / 0.82	0.85 / 0.82	0.86 / 0.87

> **Lower:** accuracy, precision, F1 · **Similar:** recall · **Higher:** AUC

Problems found in the paper

1. The KNN and SVM settings are swapped.
2. The SVM gamma (0.65) does not fit the 0–10 scaling.
3. Numbers differ: Tree AUC 0.88 vs 0.86; KNN AUC 0.91 vs 0.92.
4. The split is not described (random_state, stratify).
5. Patients after cleaning and class balance not reported.

> These gaps make an exact reproduction impossible.

Suggested improvements

One-hot encoding

Categories like chest pain type have no natural order.

Standard 0–1 scaling

Works with the default SVM settings.

Repeated CV

More stable results than one 60-patient test set.

Threshold tuning

Lower the threshold to miss fewer sick patients.

More data

Add the other UCI hospitals.

Report everything

Split, random_state and scaling order, so others can reproduce it.

PART 3 · CARLOS

Conclusion

RECOMMENDED MODEL

SVM (RBF, gamma = 0.001)

Accuracy 85% · AUC 0.96 · recall 0.79

- Approximate reproduction, same level of performance.
- The paper’s SVM settings fail; tuning fixed them.
- Models differ by only 1–3 patients.
- Limits: small, old, mostly male data; not clinical.

Thank you!

Questions?

Carlos Alvarez, Felipe Kuerten Neves & Karina Adorno

Code and notebook: github.com/karinaadorno93-sudo/Midterm-Project